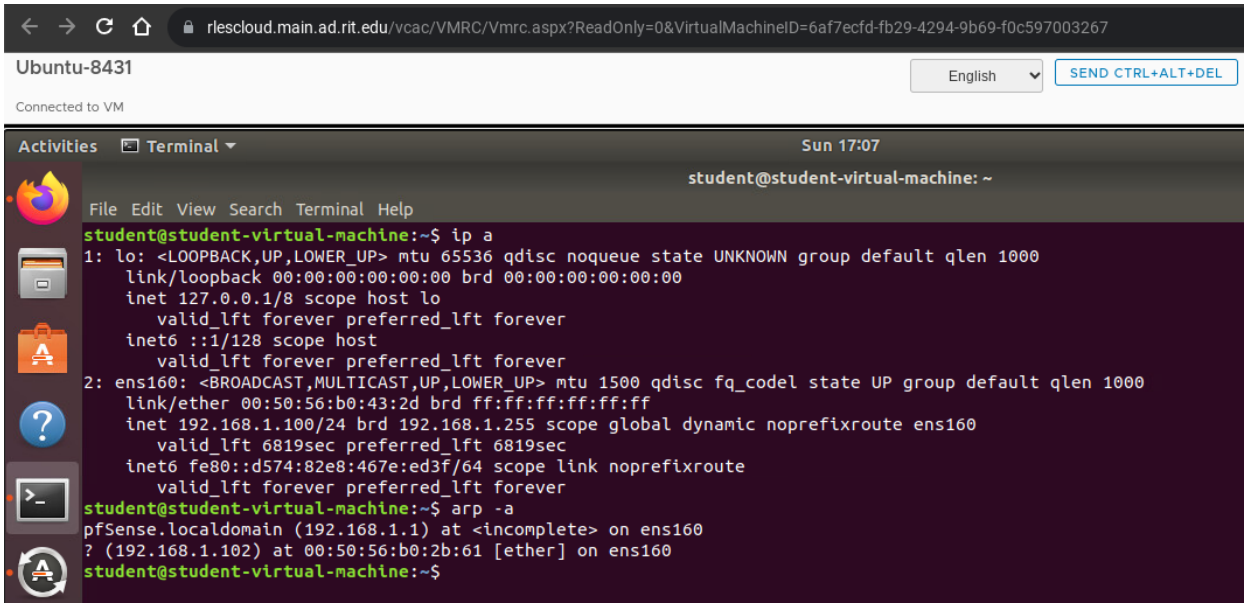
Part 1:

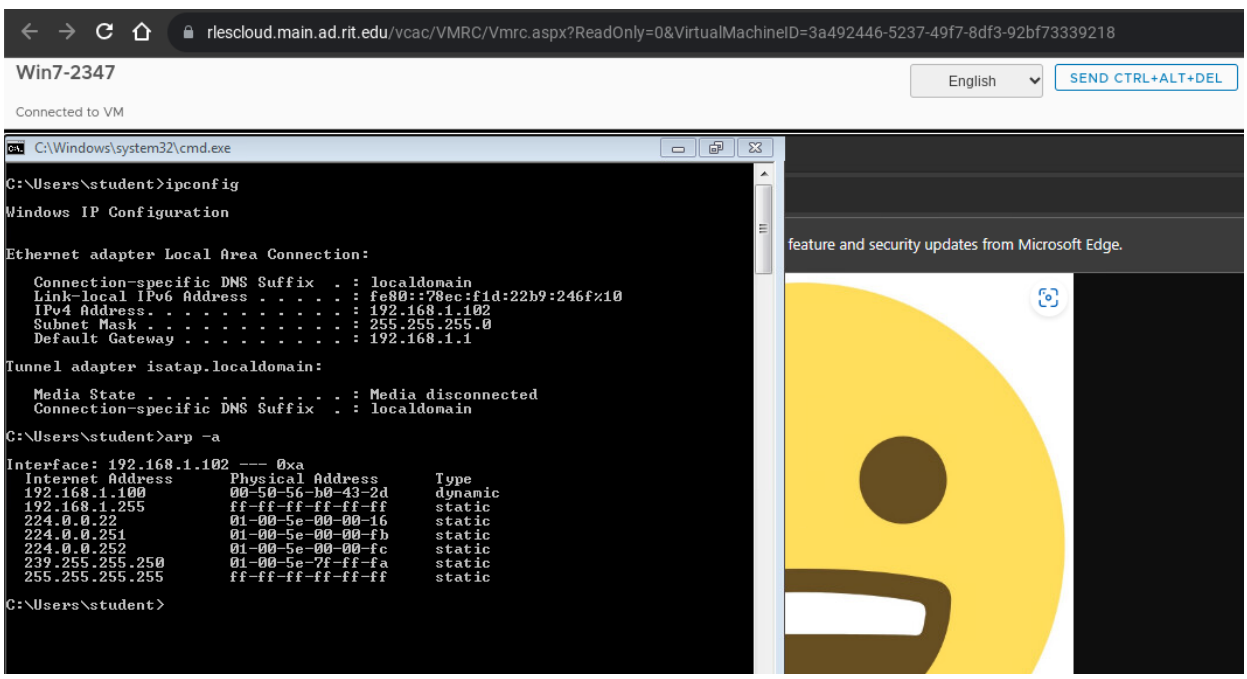
1. Lets just say A1 and A2 are connected to bus A and B1, B2, B3 are connected to bus B. Bus A and all the machines connected to it are one collision domain. Bus B and all the machines connected to it are another collision domain. The broadcast domain is the whole connection. Both bus A and all its machines, bus B and all its connections are together in one broadcast domain.
2. We deal with having more machines than addresses on the internet by using NAT for IPv4. There are internal network or private IP addresses. Machines inside the network get those private addresses. When these devices access the Internet, the router translates their private IP addresses into a single public IP address. This allows multiple devices within the private network to share a single public IP address. NAT helps extend the usability of IPv4 addresses. The way that NAT keeps track of internal connection to the internet is by using ports. So when there is outbound traffic to a certain internal machine, the NAT sees what port that machine is connected to and forwards the traffic. As for determining who the public IP address belongs to is by using the WHOIS database. It's a protocol and database that is used to look up information about IP addresses, domain names, It provides registration and contact information, etc.
3. On the host end, the application transport and network layer needs to be implemented. In the router, the network and link layer has been implemented.
4. You can use both MAC and IP blacklist to stop users from accessing your web server. MAC blocklist can be used in internal networks and IP blacklist can be used to prevent users from outside the network.
5. Yes we would still need a certificate to authenticate the site. You might always get a correct IP address for the site, but it doesn't prevent man-in-the-middle attacks.Certificates help to protect against man-in-the-middle attacks by providing a way for the client to verify the identity of the server. It provides an additional layer of security that helps to protect against man-in-the-middle attacks. Certificates are also used to encrypt traffic between the client and the server.

Part 2:

1. "Allow from all" tells the server to accept connections from everywhere.
2. Logging with the credential "admin, 4dM1n!" shows an image of the happy face emoji.
3. **Output of the arp -a**
    a. Ubuntu VM:

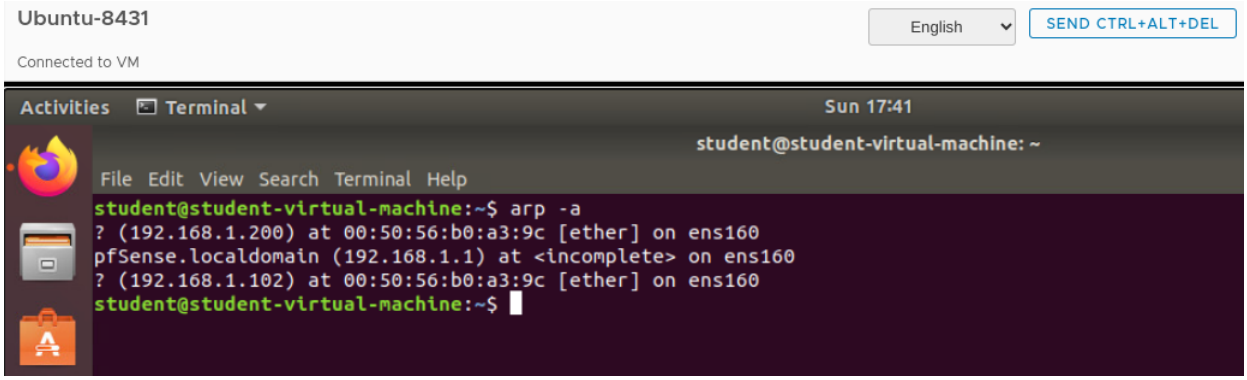


b. Windows VM:

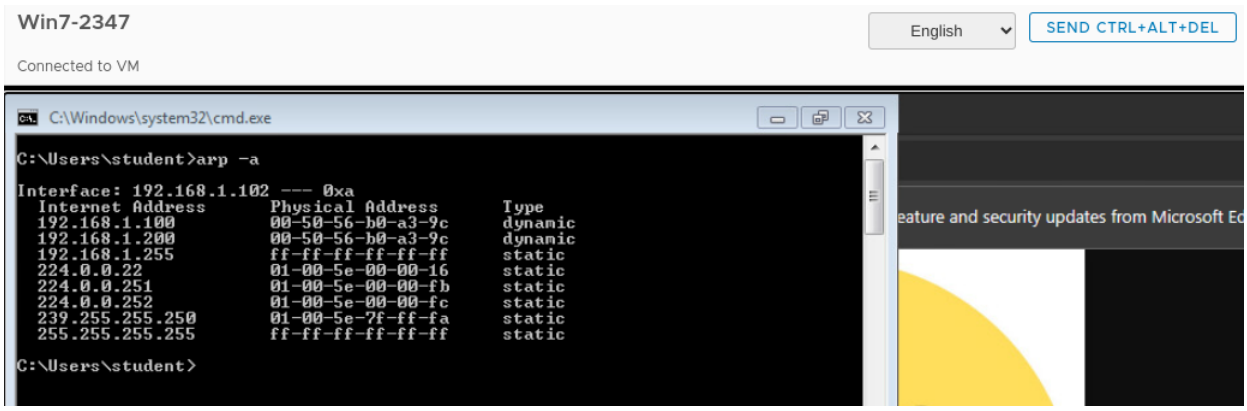
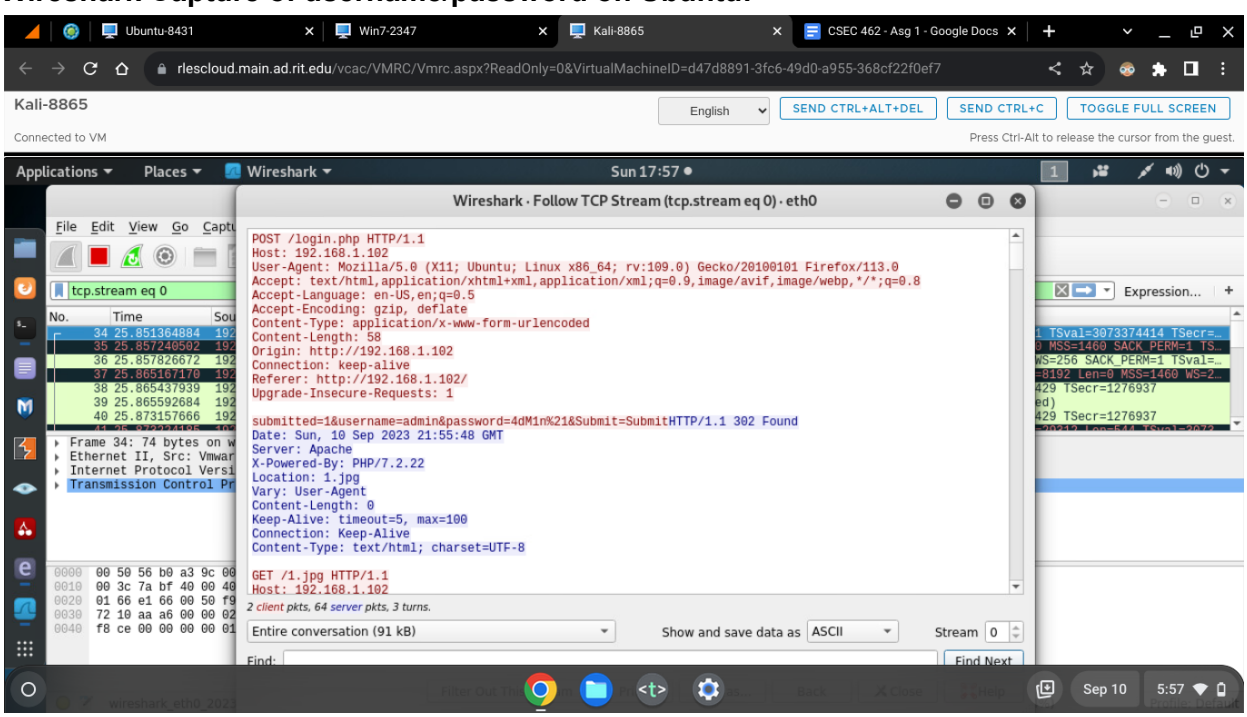


After ARP Poison, arp -a shows:

1. Ubuntu VM:



2. Windows VM:



Wireshark Capture of username/password on Ubuntu: